

Defragmenting

Regularly defragmenting the hard disk maintains optimum performance for read/write operations. Hard disks store data in sectors and clusters, the latter being the smallest addressable unit. Clusters are of a fixed size, depending on the file system (FAT, FAT32, NTFS, etc). A cluster can hold only one file, but a file may span over several clusters. For FAT32 partitions, the cluster size is 4 KB. Thus, any file between 0 bytes to 4 KB will occupy one cluster. Should its size increase beyond 4 KB, it will look for the next free cluster to fill up. With frequently changing

files such as documents, spreadsheets, images, etc, the fragments of the file may not be on contiguous clusters. Reading and writing to such files spread all over the partition is obviously slow. Defragmenting brings pieces of the file together, so that they are accessed faster.

All versions of Windows are bundled with defragmenting tools. In Windows 98, you can run it from **Start > Programs > Accessories > System Tools > Disk Defragmenter**. In Windows 2000 and XP, right-click My Computer and click Manage. Look for Disk Defragmenter under Storage.

uses a variable swap file that is created on bootup and where the size is dynamically modified when required. Though this works just fine, it is not the most optimal. You can never get the speed of RAM from the hard disk, but you can make it slightly faster. The best way to configure the swap file is to have a permanent swap file, preferably on a dedicated partition. This minimises the effect of disk fragmentation and increases seek time. Another practice that a few power users have adopted is to place the swap file on the first partition and the operating system on the following partition. Data closer to the centre of the drive is read slightly faster than data at the periphery. This is especially useful for PCs with only

64 or 128 MB of RAM where data would frequently be swapped to the hard disk and can be easily done with BIOSes that allow booting from the D drive.

To change the swap file settings, open **Control Panel > System > Performance** and click Virtual Memory. Choose 'Let me specify my own virtual memory settings' and select the drive for the swap file. Note that no matter what partition you boot from, it will always appear as the C drive from DOS and Windows 98, and the first partition that is recognised as C by the BIOS will be assigned a higher drive letter. Set the minimum and maximum size for the swap file. Both these numbers should be identical to set a permanent

(non-variable) swap file. Ideally, it should be two-and-a-half times the amount of RAM on your system, but if you have a separate partition for it, you may not want it to be less than 512 MB—if it is lower, it will have to be a FAT16 partition, not FAT32.

Avoid the swap file

When Windows loads, it moves certain parts of the operating system to the swap file anyway, regardless of whether it has run out of RAM or not. This is very good for systems with less memory, since it leaves some free RAM and applications will start much faster. Otherwise, when you start an application and memory is required, Windows will have to first move all this data to the swap file to free up RAM. This functionality is usually not required on systems with 256 MB or more RAM, especially if they only run common productivity tools such as Office and Internet applications. You can prevent Windows from using the swap file until absolutely required by adding the line below to the [386Enh] section of the \Windows\system.ini file.

ConservativeSwapfile Usage=1

You can edit this file in Notepad. In case you face any problems such as programs crashing frequently, remove this line and reboot.

Display settings

The display settings affect the speed of Windows too. The more animations and transi-

tions the OS has to show, the higher is the processor and RAM usage. Typically, turning off all such features instantaneously shows the difference in speed. Right-click an empty area of the desktop and click Properties. Under the Effects tab, uncheck the boxes for 'Animate windows, menus and lists', 'Smooth



Turn off unnecessary graphic effects

edges of screen fonts' and 'Show window contents while dragging'.

Temporary cleaner

Regularly removing files that accumulate in the Temp folder can also show better performance—these files are usually very small and unnecessarily fill up the hard disk. This also causes high disk fragmentation and pushes important data towards the periphery of the disk, where read/write operations are slower. Ideally, create a batch file that empties this folder and place it in the Startup, so that it runs every time you boot to Windows. You could do this from the autoexec.bat too, but this file runs while still in DOS mode, so disk access will be much

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